

Decent Coaching Classes

X Semi

Mind Your Math - II

Answer Key: Test - Chapters 1 and 2

Q.1 (A) Choose the correct alternative

- (1) (B) $ax + by + c = 0$
- (2) (B) no solution or infinitely many solutions
- (3) (B) $x^2 + 5x - 2 = 0$
- (4) (B) real and equal

Q.1 (B) Solve the following (any four)

- (1) $a = 2, b = -3, c = 5$
- (2) $D = (1)(-1) - (1)(1) = -1 - 1 = -2$
- (3) $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- (4) Putting $x = 2$: $(2)^2 - 5(2) + 6 = 4 - 10 + 6 = 0$. Yes, $x = 2$ is a root.
- (5) $|4 \ 3; 2 \ 5| = (4 \times 5) - (2 \times 3) = 20 - 6 = 14$

Q.2 Activities (any two)

- (1) $D = (2 \times -2) - (1 \times 3) = -4 - 3 = -7$
 $Dx = (8 \times -2) - (-3 \times 3) = -16 - (-9) = -7$
 $Dy = (2 \times -3) - (1 \times 8) = -6 - 8 = -14$
 $x = Dx \div D = -7 \div -7 = 1, y = Dy \div D = -14 \div -7 = 2 \therefore (x, y) = (1, 2)$
- (2) $k(4) - 14 + 6 = 0 \therefore 4k - 14 + 6 = 0 \therefore 4k - 8 = 0 \therefore 4k = 8 \therefore k = 2$
- (3) $x^2 - 9x + 18 = x^2 - 6x - 3x + 18 = x(x - 6) - 3(x - 6) = (x - 6)(x - 3)$

Q.3 Solve the following (any four)

- (1) Adding the equations: $2x = 10 \Rightarrow x = 5$. Substituting: $5 + y = 7 \Rightarrow y = 2. \therefore (x, y) = (5, 2)$
- (2) $D = (2)(-1) - (1)(1) = -3$; $Dx = (7)(-1) - (2)(1) = -9$; $Dy = (2)(2) - (1)(7) = -3$
 $x = Dx/D = -9/-3 = 3, y = Dy/D = -3/-3 = 1 \therefore (x, y) = (3, 1)$
- (3) $x^2 - 7x + 12 = (x - 3)(x - 4) = 0 \therefore x = 3$ or $x = 4$
- (4) $x^2 + 2x - 15 = (x + 5)(x - 3) = 0 \therefore x = -5$ or $x = 3$
- (5) $\Delta = (-4)^2 - 4(1)(4) = 16 - 16 = 0$. Since $\Delta = 0$, the roots are real and equal.
- (6) $a = 1, b = -5, c = 6. x = \frac{5 \pm \sqrt{25 - 24}}{2} = \frac{5 \pm 1}{2} \therefore x = 3$ or $x = 2$

Q.4 Solve the following (any two)

- (1) $D = (1)(-2) - (3)(2) = -2 - 6 = -8$
 $Dx = (5)(-2) - (7)(2) = -10 - 14 = -24$
 $Dy = (1)(7) - (3)(5) = 7 - 15 = -8$
 $x = Dx/D = -24/-8 = 3, y = Dy/D = -8/-8 = 1 \therefore (x, y) = (3, 1)$
- (2) Factorisation: $x^2 + 3x - 4 = (x + 4)(x - 1) = 0 \Rightarrow x = -4$ or $x = 1$
Completing the square: $x^2 + 3x = 4 \Rightarrow x^2 + 3x + 2.25 = 6.25 \Rightarrow (x + 1.5)^2 = 6.25 \Rightarrow x + 1.5 = \pm 2.5 \Rightarrow x = 1$ or $x = -4$
Formula: $a = 1, b = 3, c = -4. x = \frac{-3 \pm \sqrt{9 + 16}}{2} = \frac{-3 \pm 5}{2} \Rightarrow x = 1$ or $x = -4$
All three methods give the same roots: $x = 1$ and $x = -4$.
- (3) Let the unit's digit = x and ten's digit = y . Number = $10y + x$; number with digits interchanged = $10x + y$.
Sum: $(10y + x) + (10x + y) = 99 \Rightarrow 11x + 11y = 99 \Rightarrow x + y = 9 \dots(I)$

Given: $y = x + 3$...(II). Substituting in (I): $x + (x + 3) = 9 \Rightarrow 2x = 6 \Rightarrow x = 3, y = 6$

$\therefore$ The number = $10y + x = 10(6) + 3 = 63$

(4) Let Rohan's present age = x years. Age 5 years ago = $(x - 5)$; age 3 years hence = $(x + 3)$.

Given: $(x - 5)(x + 3) = 84 \Rightarrow x^2 - 2x - 15 = 84 \Rightarrow x^2 - 2x - 99 = 0$

$\Delta = (-2)^2 - 4(1)(-99) = 4 + 396 = 400, \sqrt{400} = 20$

$x = [2 \pm 20] / 2 \Rightarrow x = 11$ or $x = -9$ (rejected, age cannot be negative)

$\therefore$ Rohan's present age = 11 years